IP and Domain Threat Intel

Turn a raw domain into a contextual artefact:

- who owns it
- what IPs it resolves to
- how often it changes
- whether it Behaves like a normal content delivery network or a throwaway setup

Tools we gotta use: nslookup.io or dnschecker.org

Core DNS Records for Triage

these are the records that matters most:

- **A/AAAA Records**: Map the domain to IPv4 and IPv6 addresses
  - whether the records hop Between different networks or not

- **NS Records**: Identify the nameservers controlling the domain

- **MX Records**: Define which servers handle email.

- **TXT Records**: Store SPF and DKIM rules or verification tags
  - Basically it tells whether the email is coming from the right person and ~~one~~ no one is intrupting in Between.

- **SOA Record**: Points to the zone's primary authority and often includes contact information

- **TTL (Time to Live)**: Tells how long to cache answers

Attack techniques using DNS:

- Fast Flux Hosting: they rotate many IPs quickly with short cache times to avoid simple Blocks.

- CDN abuse: Legitimate CDNs like Cloudflare or Akamai change IPs too, but do~~ne~~ within their ASN ecosystem.

- Typosquatting: visual trick

- IDN (Internationalised domain name): creating look-alike domains.

SOC Analyst workflow:

- Snap shot current DNS
- Basic ownership check
- Interpret Patterns
- Log Evidence
- Recommend Action

IP Enrichment

Enrichment: is adding ownership, ASN (Autonomous System Number), geolocation, and service context to an IP so that our decision is evidence-driven. &

Main Idea : An IP address by itself does not tell us enough

That IP could belong to anyone. So before reacting, the analyst must enrich the IP. That means adding extra information so the decision is based on evidence, not guesswork.

What IP enrichment means:
- it is collecting more context about an IP address,

This helps avoid mistakes like:
- overreacting : blocking a legitimate
- Undersreacting : ignoring an actual c2 server

The role of RDAP

RDAP → Registration Data Access Protocol

It is used to look up the official registration details for an IP address

RDAP is like an authoritative ownership database

From RDAP we can get:
- Netrange : the range of addresses delegated
- Organisation: the registred holder
- Remarks : Often include whether the block is used for hosting, broadband, or mobile
- Abuse Contact: The official mailbox for incident reporting

ASN and why it matters:

ASN = Autonomous System Number

ASN represents a network or group of IP ranges collected By one organization for internet routing

Basically, ASN tells what larger network this IP Belongs to

Hosting ASN: ASN hosting usually used by data centers and web hosting providers
- That means attackers often rent servers there
- and suspicious domains are commonly hosted there

Residential ISPs : If they ~~come from~~ attacks belongs from home internet
- that means home is hacked

Cloud/CDN ASNs : Global anycast, dozens of prefixes
- Blocking whole range here causes collateral damage.

- SOC Analyst Workflow

- Start with RDAP → Check netrange, org, ASN and abuse contact
- Add ASN Context → Check bgpview.io or ipinfo.io for ASN details
- Check Geolocation
- Look for rDNS Patterns
- Consult Internal Logs → has this IP appeared in the last 30 days?

- Classify Role
- Plan Outreach

nslookup
whois

iplocation.net/ip-lookup → helps us the see the region
rdap.org → gives info about ip
ipinfo.io → gives us ASN-number

shodan.io → gives us the domains, open ports, and certifications

crt.sh → gives us all the certifications

censys.io → shows exposed services even on non-standard ports ~~and~~ which was so amazing
platform

We also need to know if the IP or domain have done malicious activity.

Even IP and Domain may change, hash doesnot so we use ~~3~~ VirusTotal, Cisco Talos Intelligence

Another great tool is IP2Proxy → which tells if the IP is using VPN or Proxy, or not.